NLP-Based Classification of Schizophrenia Using the DAIS-C Corpus

Formal thought disorder (FTD)—marked by disorganized, incoherent, or tangential speech—is a core diagnostic feature of schizophrenia (Stein et al., 2025) and is associated with higher relapse rates, longer hospitalizations, and persistent social impairment (Tan & Rossell, 2019). Yet clinical evaluation of FTD still depends largely on qualitative interviews and rater-based scales, leaving assessments vulnerable to subjectivity and variability. Recent advances in Natural Language Processing (NLP) offer opportunities to quantify linguistic disruptions objectively. Building on findings that psychosis- and FTD-related speech shows reduced semantic structure and increased lexical unpredictability (Li et al., 2025; Salman et al., 2025), this study evaluates the diagnostic potential of combining long-context transformer representations with token-level surprisal, capturing both discourse-level coherence and local lexical predictability.

We analyzed transcripts from the DAIS-C corpus (Delgaram-Nejad et al., 2023), consisting of semi-naturalistic speech from individuals with schizophrenia and matched comparison speakers (15 clinical, 13 comparison). To examine the role of context length, we segmented each transcript into 14 dataset splits ranging from 1 to 14 sentences, the maximum size permitted by GPT-Neo's context window. Long-range discourse disruptions—often described as failures to maintain a coherent discourse plan (Barch & Berenbaum, 1997)—were modeled using Longformer (Beltagy et al., 2020), which processes up to 4,096 tokens via sliding-window attention. Token-level surprisal values ($-\log P(w_t|w_1,\ldots,w_{t-1})$) were extracted using GPT-Neo and aggregated into global metrics (Mean, Variance, Slope). To further isolate discourse anomalies, we computed the mean (Top10_Avg) and maximum (Top10_Max) Longformer embeddings corresponding to the top 10% highest-surprisal tokens. Model evaluation was conducted using 100 bootstrap iterations; in each iteration, the resampled dataset was divided into 80% training and 20% test sets.

Results revealed a clear hierarchy across models and feature sets (Fig. 1–3). Surprisal-only feature sets yielded the lowest Accuracy, F1 score, and ROC_AUC values, consistent with the substantial overlap observed in the Top-10% surprisal distributions across groups (Fig. 4–5). In contrast, all Longformer-based feature sets exceeded .90 across all metrics, and even the Top-10% embedding vectors alone consistently outperformed the surprisal-only configurations. Across classifiers, multilayer perceptron (MLP) models reliably outperformed logistic regression (LR). Notably, the MLP model using the Longformer_CLS+Surprisal feature set for the 13-sentence split achieved the strongest overall performance, with Accuracy = 0.9667, F1 = 0.9755, and ROC_AUC = 0.9973.

These findings demonstrate that Longformer-based models, which encode extended contextual structure, reliably detect linguistic disruptions associated with FTD. While token-level surprisal provides meaningful local predictability signals, it becomes diagnostically informative only when integrated with long-range contextual representations. This pattern aligns with the view that FTD affects multiple layers of linguistic structure, with discourse coherence playing a central role (Barch & Berenbaum, 1997), and underscores the need for frameworks that incorporate global predictability measures such as perplexity—$\exp(-(1/N) \times \Sigma \log P(w_i))$—which quantifies a model's uncertainty over an entire sequence and has been proposed as a sensitive marker of disorganized or psychosis-related language (Li et al., 2025).

(491 words)

Fig. 1. *Accuracy Across Models*

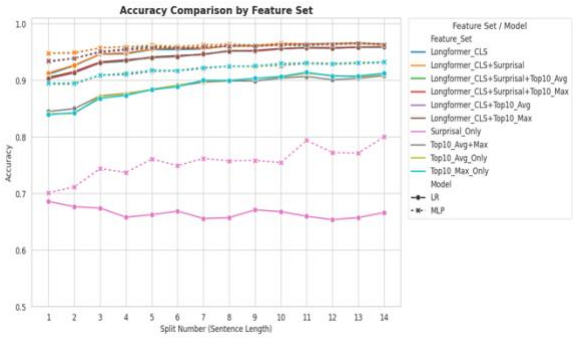

Fig. 2. *F1 Scores Across Models*

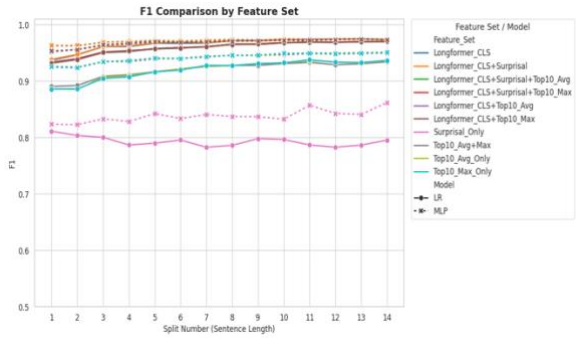

Fig. 3. *ROC_AUC Across Models*

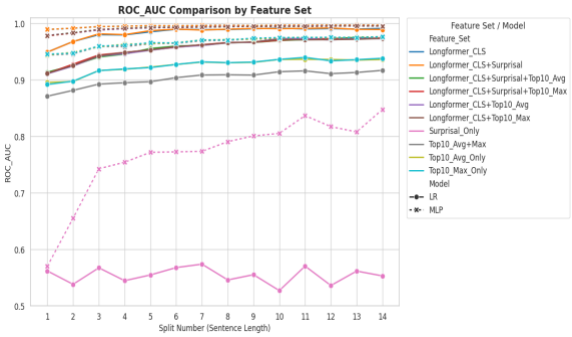

Fig. 4. *Surprisal-Related Metrics Comparison*

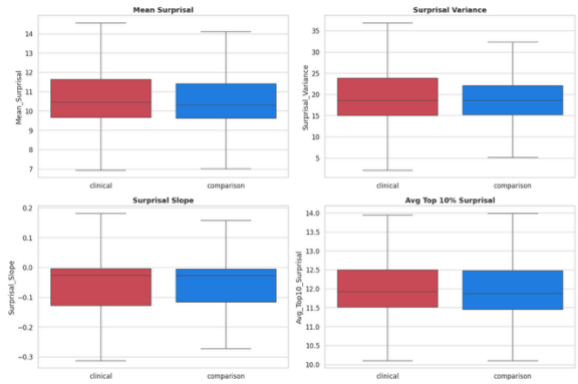

Fig. 5. *Example of Top 10% High-Surprisal Words: Clinical vs. Comparison Group Text*

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